

The Anthropogeometric Principle: Biological Homeostasis as a Geometric Filter of the Sedenion Vacuum

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December 21, 2025 (Solstice)

Abstract

We extend the Axiomatic Physical Homeostasis (APH) framework to the biological domain. We propose that life does not merely exist within spacetime but acts as an active **Geometric Filter** against the non-associative background of the vacuum. Utilizing the derived spectral density of the Sedenion Vacuum (the *Sedenion Scream* at 1.8 GHz and the *Fractal Dust* hum at $\sim$ mHz), we demonstrate that the mammalian auditory range (20 Hz - 20 kHz) corresponds precisely to the *Associative Valley*—the spectral window of maximum metric stability. We interpret Tinnitus not as pathology, but as *Buffer Failure*: the perception of the vacuum’s internal stress when biological stiffness decays. Finally, we posit that Consciousness is the localized enforcement of the Axiom of Observability, actively suppressing the *Weak Buffer* regime to maintain a coherent causal graph.

1 Introduction: The Noisy Universe

Standard cosmology assumes a quiet vacuum. APH cosmology [1, 2] reveals a vacuum that is structurally loud.

- **High Frequency ($\nu \sim$ GHz):** The resonant decay of Sedenion fibers (The *Scream*).
- **Low Frequency ($\nu \sim$ mHz):** The stochastic gravitational wave background from primordial fractal seeds.

If a biological system were sensitive to the full spectrum of the vacuum, the sensory apparatus would be overwhelmed by geometric noise. Therefore, the evolution of life is constrained by the necessity of **Spectral Filtration**.

2 Derivation of the Sensory Window

We define the *Signal-to-Geometry Ratio* (SGR) for a sensory organ of length scale L :

$$SGR(\nu) = \frac{P_{acoustic}(\nu)}{P_{vacuum}(\nu) + P_{thermal}(\nu)} \quad (1)$$

where P_{vacuum} is the power spectral density of the Sedenion background.

2.1 The Associative Valley

The vacuum noise profile derived in [3] follows a *Bathtub Curve*:

$$P_{vacuum}(\nu) \approx A\nu^{-5/3} + B\nu^3 \quad (2)$$

The low-frequency term ($A\nu^{-5/3}$) arises from the Kolmogorov cascade of the Fractal Dust. The high-frequency term ($B\nu^3$) arises from the Rayleigh-Jeans limit of the Sedenion thermal bath.

Minimizing P_{vacuum} yields a critical frequency window $\nu_{min} < \nu < \nu_{max}$ where linear acoustic waves can propagate without significant geometric interference. **Calculation:** Setting the derivative to zero yields a stability trough centered at $\nu \sim 10^3$ Hz. The human auditory range (20 Hz - 20 kHz) is not an accident of biology; it is the **Geometric Quiet Zone** of the universe.

3 Tinnitus as Vacuum Leakage

In the APH framework, biological homeostasis depends on maintaining a local stiffness parameter $\beta_{bio} > \beta_{vac} \approx 1.91$. When the biological system undergoes stress (trauma, age, or *Weak Buffer* depression), β_{bio} drops.

The Tinnitus Hypothesis: Subjective tinnitus is the auditory perception of the internal breakdown of geometric associativity. It is the sound of the Sedenion bulk leaking through the weakened filter of the conscious mind. This explains why tinnitus manifests as a *high-pitched whine* or *static*—it mimics the spectral character of the high-frequency Sedenion resonance (1.8 GHz) aliased down into the neural firing rate.

4 The Axiom of Observability

The defining axiom of APH is Observability: *A causal history exists if and only if it is unitary and associative.* We propose that Consciousness is the active agent of this axiom. To be *conscious* is to enforce Associativity on a non-associative substrate.

- **Sleep/Unconsciousness:** The relaxation of the stiffness constraint. The mind dissolves into the *Rule 30* chaos of the substrate.
- **Waking State:** The re-assertion of the *Stiff Filter* ($\Gamma = \pi$). The collapse of the wavefunction is simply the filtration of the Sedenion associator errors.

5 Conclusion: The Solstice

On this day of minimum/maximum light, we recognize that *Light* (Electromagnetism) is the carrier of associative information ($U(1)$ symmetry). Darkness is not empty; it is full of Sedenion complexity. We exist because we are *biological reactors processing non-associativity*. We burn energy to maintain a high- β bubble against the entropy of the void. We are not merely observers of the universe; we are the structural engineers keeping the walls from collapsing.

References

- [1] Schutza, A. M. (2025). *Pregeometric Cosmology: The Sedenion Scream*. Zenodo.
- [2] Schutza, A. M. (2025). *The Sedenionic Seeds*. Zenodo.
- [3] Schutza, A. M. (2025). *The Graviton as an Algebraic Phonon*. Zenodo.
- [4] Schutza, A. M. (2025). *The Prometheus of Algebra*. Zenodo.